

# General Biology Complete Notes

Cell • Digestion • Respiration • Excretion • Skin • Brain

## DEPARTMENT

Mathematics

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## 1. Introduction to Biology

### Definition — Biology

**Biology** is the branch of science that deals with the study of living organisms — their structure, function, growth, origin, evolution, and distribution.

### Branches of Biology

| Branch       | Study of                            |
|--------------|-------------------------------------|
| Zoology      | Animals                             |
| Botany       | Plants                              |
| Microbiology | Microorganisms                      |
| Genetics     | Heredity and variation              |
| Ecology      | Organisms and their environment     |
| Physiology   | Functions of organs and systems     |
| Anatomy      | Structure of organisms              |
| Biochemistry | Chemical processes in living things |

### Levels of Organization

Atom → Molecule → Organelle → Cell → Tissue → Organ → Organ System → Organism → Population → Ecosystem

### Characteristics of Living Things

- Cellular organization
- Metabolism
- Growth and development
- Reproduction
- Response to stimuli
- Homeostasis
- Adaptation
- Nutrition

## 2. Cell and Their Organelles

### Definition — Cell

The **cell** is the basic structural and functional unit of life. All living organisms are made of one or more cells. Proposed by **Robert Hooke** (1665).

## Types of Cells

- **Prokaryotic Cell:** No true membrane-bound nucleus. No membrane-bound organelles. e.g. Bacteria, Blue-green algae.
- **Eukaryotic Cell:** Has a true nucleus enclosed in nuclear membrane. Has membrane-bound organelles. e.g. Plant cells, Animal cells.

## Cell Organelles and Their Functions

| Organelle                  | Function                                                  |
|----------------------------|-----------------------------------------------------------|
| Cell Membrane              | Controls movement of substances in/out of cell            |
| Cell Wall (plants)         | Provides rigidity and support; made of cellulose          |
| Nucleus                    | Controls all cell activities; contains DNA                |
| Nucleolus                  | Synthesizes ribosomal RNA (rRNA)                          |
| Mitochondria               | Site of cellular respiration; produces ATP (“powerhouse”) |
| Ribosome                   | Site of protein synthesis                                 |
| Endoplasmic Reticulum (ER) | Rough ER: protein synthesis; Smooth ER: lipid synthesis   |
| Golgi Apparatus            | Packages and secretes proteins (“post office”)            |
| Lysosome                   | Contains digestive enzymes; destroys waste                |
| Vacuole                    | Storage of water, food, waste                             |
| Chloroplast (plants)       | Site of photosynthesis; contains chlorophyll              |
| Centrosome                 | Forms spindle fibers during cell division                 |
| Cytoplasm                  | Jelly-like fluid; medium for chemical reactions           |

### Key Point — Cell Division

**Mitosis:** Cell divides to produce 2 identical daughter cells (growth, repair). Phases: Prophase → Metaphase → Anaphase → Telophase.

**Meiosis:** Cell divides to produce 4 genetically different cells with half the chromosomes (sexual reproduction).

## 3. Digestion System

**Definition — Digestion**

**Digestion** is the process of breaking down complex food molecules into simpler absorbable forms using mechanical and chemical processes.

**Organs of the Digestive System****Alimentary Canal (in order):**

Mouth → Pharynx → Oesophagus → Stomach → Small Intestine → Large Intestine → Rectum → Anus

| Organ           | Function                                                       |
|-----------------|----------------------------------------------------------------|
| Mouth           | Mechanical digestion (chewing); salivary amylase breaks starch |
| Oesophagus      | Transports food to stomach via peristalsis                     |
| Stomach         | Churns food; HCl kills bacteria; pepsin digests proteins       |
| Small Intestine | Main site of digestion and absorption of nutrients             |
| Liver           | Produces bile (emulsifies fats); detoxification                |
| Pancreas        | Produces digestive enzymes (amylase, lipase, protease)         |
| Large Intestine | Absorbs water; forms faeces                                    |
| Rectum / Anus   | Storage and elimination of undigested waste                    |

**Digestive Enzymes**

| Enzyme           | Source          | Substrate | Product                |
|------------------|-----------------|-----------|------------------------|
| Salivary Amylase | Salivary glands | Starch    | Maltose                |
| Pepsin           | Stomach         | Proteins  | Peptides               |
| Lipase           | Pancreas        | Fats      | Fatty acids + Glycerol |
| Trypsin          | Pancreas        | Proteins  | Amino acids            |
| Maltase          | Small intestine | Maltose   | Glucose                |

**Key Point — Absorption**

Nutrients are absorbed in the **small intestine** through finger-like projections called **villi** (singular: villus). Each villus has microvilli (brush border) to increase surface area. Glucose and amino acids pass into blood capillaries; fatty acids into lacteals (lymph vessels).

## 4. Respiratory System

**Definition — Respiration**

**Respiration** is the process by which organisms exchange gases with their environment and release energy from organic molecules.

### Types of Respiration

- **Aerobic Respiration:** Uses oxygen; produces  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ , and energy (ATP).



- **Anaerobic Respiration:** No oxygen; less ATP produced.

In muscles: Glucose  $\rightarrow$  Lactic acid + 2 ATP

In yeast: Glucose  $\rightarrow$  Ethanol +  $\text{CO}_2$  + 2 ATP

### Organs of the Respiratory System

**Pathway of air:**

Nose  $\rightarrow$  Pharynx  $\rightarrow$  Larynx  $\rightarrow$  Trachea  $\rightarrow$  Bronchi  $\rightarrow$  Bronchioles  $\rightarrow$  Alveoli

| Structure   | Function                                                                  |
|-------------|---------------------------------------------------------------------------|
| Nose        | Filters, warms, and moistens air                                          |
| Trachea     | Carries air to lungs; supported by C-shaped cartilage rings               |
| Bronchi     | Two branches leading to left and right lungs                              |
| Bronchioles | Smaller branches within lungs                                             |
| Alveoli     | Tiny air sacs; site of gas exchange ( $\text{O}_2$ in, $\text{CO}_2$ out) |
| Diaphragm   | Muscular sheet; controls breathing by changing lung volume                |

### Gas Exchange at Alveoli

- Alveoli are richly supplied with blood capillaries

- $O_2$  diffuses from alveoli into blood (high  $\rightarrow$  low concentration)
- $CO_2$  diffuses from blood into alveoli
- Features of alveoli: large surface area, moist thin walls, rich blood supply

### Breathing Mechanism

|              | Inhalation           | Exhalation         |
|--------------|----------------------|--------------------|
| Diaphragm    | Contracts (flattens) | Relaxes (domes up) |
| Rib cage     | Moves up and out     | Moves down and in  |
| Lung volume  | Increases            | Decreases          |
| Air pressure | Decreases            | Increases          |
| Air flow     | Into lungs           | Out of lungs       |

## 5. Excretion System

### Definition — Excretion

**Excretion** is the removal of metabolic waste products from the body. These are substances produced during chemical reactions in cells that would be toxic if allowed to accumulate.

### Excretory Organs and Their Waste Products

| Organ   | Waste Excreted            | Method        |
|---------|---------------------------|---------------|
| Kidneys | Urea, excess water, salts | Urine         |
| Lungs   | $CO_2$ , water vapour     | Exhalation    |
| Skin    | Water, salts, small urea  | Sweat         |
| Liver   | Bile pigments (bilirubin) | Bile into gut |

### The Kidney

- Humans have two kidneys located in the abdominal cavity
- Functional unit of kidney: **Nephron**
- Each kidney contains approximately 1 million nephrons

## Nephron and Urine Formation

**Process:** Filtration → Reabsorption → Secretion → Urine formation

**Steps:**

1. **Glomerular Filtration:** Blood filtered under pressure in Bowman's capsule; glucose, water, urea, salts pass into tubule
2. **Tubular Reabsorption:** Useful substances (glucose, water, amino acids) reabsorbed back into blood
3. **Tubular Secretion:** Additional wastes ( $H^+$ ,  $K^+$ ) secreted into tubule
4. **Urine:** Remaining fluid (urea, excess salts, water) passes to ureter → bladder → urethra

### Key Point — Osmoregulation

The kidneys regulate water balance of the body (osmoregulation) under the control of **ADH** (Anti-Diuretic Hormone) secreted by the pituitary gland. High ADH → more water reabsorbed → concentrated urine.

## 6. Skin

### Definition — Skin

The **skin** is the largest organ of the body. It forms the outer covering and serves as the primary barrier against the environment.

### Layers of the Skin

- **Epidermis** (outer layer):
  - Outermost layer: dead cornified cells (keratin)
  - Contains melanocytes (produce melanin — skin pigment)
  - No blood vessels; cells replaced continuously
- **Dermis** (inner layer):
  - Contains hair follicles, sweat glands, sebaceous (oil) glands
  - Rich in blood vessels, nerve endings, lymph vessels
  - Contains collagen and elastin fibres
- **Hypodermis** (subcutaneous layer):
  - Fat layer; insulation and energy storage

## Functions of the Skin

| Function               | Details                                                         |
|------------------------|-----------------------------------------------------------------|
| Protection             | Barrier against pathogens, UV radiation, chemicals, dehydration |
| Temperature regulation | Sweating cools body; blood vessels dilate/constrict             |
| Excretion              | Removes water, salts, and small amounts of urea through sweat   |
| Sensation              | Nerve endings detect touch, pain, pressure, temperature         |
| Vitamin D synthesis    | UV light converts cholesterol to Vitamin D                      |
| Waterproofing          | Keratin prevents excessive water loss                           |

### Key Point — Thermoregulation

When body is **too hot**: blood vessels dilate (vasodilation), sweat glands produce sweat, hair lies flat → body cools.

When body is **too cold**: blood vessels constrict (vasoconstriction), shivering generates heat, hair stands up (goosebumps) → body warms.

## 7. The Brain

### Definition — Brain

The **brain** is the main organ of the central nervous system (CNS), located in the cranium (skull). It controls all voluntary and involuntary activities of the body.



## Main Parts of the Brain

| Part              | Functions                                                                      |
|-------------------|--------------------------------------------------------------------------------|
| Cerebrum          | Largest part; controls thinking, memory, speech, voluntary movement, sensation |
| Cerebellum        | Coordinates movement, balance, and posture                                     |
| Medulla Oblongata | Controls involuntary actions: breathing, heartbeat, blood pressure, swallowing |
| Hypothalamus      | Regulates body temperature, hunger, thirst, sleep, emotional responses         |
| Thalamus          | Relay station for sensory information to cerebrum                              |
| Pituitary Gland   | Master endocrine gland; controls other glands; attached to hypothalamus        |
| Brain Stem        | Connects brain to spinal cord; controls basic life functions                   |

## Lobes of the Cerebrum

- **Frontal lobe:** Personality, decision-making, voluntary movement, speech (Broca's area)
- **Parietal lobe:** Processes touch, temperature, pain, spatial awareness
- **Temporal lobe:** Hearing, language comprehension (Wernicke's area), memory
- **Occipital lobe:** Vision and visual processing

## Neuron — Basic Unit of Nervous System

**Parts of a neuron:** Cell body (soma) → Dendrites (receive signals) → Axon (transmits signals) → Synaptic terminals

**Types:** Sensory neurons (carry signals TO brain), Motor neurons (carry signals FROM brain), Interneurons (connect sensory and motor).

### Key Point — Reflex Arc

A **reflex action** is an automatic, involuntary response to a stimulus. It does not involve the brain.

Pathway: Stimulus → Receptor → Sensory neuron → Spinal cord → Motor neuron → Effector → Response

## Quick Reference Cheat Sheet

## GENERAL BIOLOGY — MASTER SUMMARY

All key topics at a glance

### Systems Summary

|             |                                                 |
|-------------|-------------------------------------------------|
| Digestion   | Mouth → Stomach → Intestine                     |
| Absorption  | Villi in small intestine                        |
| Respiration | Alveoli: O <sub>2</sub> in, CO <sub>2</sub> out |
| Aerobic     | Glucose + O <sub>2</sub> → 38 ATP               |
| Anaerobic   | Glucose → Lactic acid + 2 ATP                   |
| Excretion   | Kidneys: urea; Lungs: CO <sub>2</sub>           |
| Nephron     | Filtration → Reabsorption                       |
| ADH         | Controls water reabsorption                     |

### Cell Organelles

|              |                      |
|--------------|----------------------|
| Nucleus      | Control centre (DNA) |
| Mitochondria | ATP production       |
| Ribosome     | Protein synthesis    |
| Golgi        | Packaging proteins   |
| Lysosome     | Digests waste        |
| Chloroplast  | Photosynthesis       |
| Cell wall    | Support (plants)     |
| Vacuole      | Storage              |

### Brain Parts & Skin Layers

#### Brain:

- Cerebrum: thinking, memory
- Cerebellum: balance
- Medulla: heartbeat, breathing
- Hypothalamus: temperature, hunger
- Thalamus: sensory relay

#### Skin Layers:

- Epidermis: outer, keratin, melanin
- Dermis: hair, sweat glands, nerves
- Hypodermis: fat, insulation

#### Skin Functions:

Protection, excretion, temperature regulation, sensation, Vitamin D

For feedback or to request additional topics:

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